



Fractal Neural Network: A new ensemble of fractal geometry and convolutional neural networks for the classification of histology images

Guilherme Freire Roberto ^{a,*}, Alessandra Lumini ^b, Leandro Alves Neves ^c,
Marcelo Zanchetta do Nascimento ^a

^a Faculty of Computer Science (FACOM) - Federal University of Uberlândia (UFU), Av. João Naves de Ávila 2121, BLB, 38400-902, Uberlândia, MG, Brazil

^b Department of Computer Science and Engineering (DISI) - University of Bologna, Via dell'Università 50, 47521, Cesena, FC, Italy

^c Department of Computer Science and Statistics (DCCE), São Paulo State University (UNESP), R. Cristóvão Colombo, 2265, 15054-000, São José do Rio Preto, SP, Brazil

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ABSTRACT

Classification of histology images is a task that has been widely explored on recent computer vision researches. The most studied approach for this task has been the application of deep learning through a convolutional neural network (CNN) model. However, the use of CNNs in the context of histological images classification has yet some limitations such as the need of large datasets, the slow training time and the difficult to implement a generalized model able to classify different types of histology tissues. In this paper, we propose an ensemble model based on handcrafted fractal features and deep learning that consists of combining the classification of two CNNs by applying the sum rule. We apply feature extraction to obtain 300 fractal features from different histological datasets. These features are reshaped into a $10 \times 10 \times 3$ matrix to compose an artificial image that is given as input to the first CNN. The second CNN model receives as input the correspondent original image. After combining the results of both CNNs, accuracies that range from 89.66% up to 99.62% were obtained from five different datasets. Moreover, our model was able to classify images from datasets with imbalanced classes, without the need for images having the same resolution, and in relative fast training time. We also verified that the obtained results are compatible with the most recent and relevant studies recently published in the context of histology image classification.

1. Introduction

Advanced cancer is typically manifested by the disordered growth of malignant tumours. There are several types of cancer, among which breast, colorectal and lymphomas three of the most common ones, and have resulted in higher mortality rates (Noone et al., 2018).

Breast cancer is a disease that initially begins with a tumour in the breast area but can later spread to other organs. Colorectal cancer consists of the growth of malignant polyps in the colon or rectum area. Lymphomas are a type of cancer that affects cells of the immunological system, wherein the most common occurrence is the non-Hodgkin lymphoma (NHL). NHLs are divided into categories, each one requiring specific treatments. According to the National Cancer Institute, the combined amount of new cases expected for these three types of cancer in 2020 is approximately 500,000 only in the United States (Howlader et al., 2017). The early diagnosis of these types of cancer is essential to provide adequate treatment and prevent death. Therefore, the

histopathological analysis is inevitable for detecting the disease (Jothi & Rajam, 2017).

Histopathology, which consists of the analysis of histological tissues and the study of how diseases affect the cells, is often performed during or after a tumour removal surgery. The tumour characteristics may determine the following treatment decisions. Usually, a pathologist performs this analysis by observing histology slides through a digital microscope (Bentaieb & Hamarneh, 2018). However, this task is prone to errors as the evaluation is often subjective and dependent on the pathologist's experience, which may lead to misdiagnosis (Mueller et al., 2016).

To provide support to pathologists, several computer vision techniques have been applied to images obtained from histology slides. These techniques consist of performing a series of evaluations on the input images and then provide a classification based on predefined classes, such as benign or malignant. This is a complex procedure, often referred as computer-aided diagnosis (CAD), which can

* Corresponding author.

E-mail addresses: guilhermefreire@gmail.com (G.F. Roberto), alessandra.lumini@unibo.it (A. Lumini), leandro@ibilce.unesp.br (L.A. Neves), marcelo.zanchetta@gmail.com (M.Z. do Nascimento).

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be split into several stages, from image acquisition, going through pre-processing, segmentation, feature extraction, feature selection and classification (Jothi & Rajam, 2017). Therefore, a CAD system is an important tool that provides a second view to the pathologist, increasing the diagnosis accuracy and reducing the amount of time and physicians required to label large amounts of medical exams (Huang et al., 2018). In this paper, we focus on the feature extraction and classification stages of a CAD system for histological image analysis.

Different techniques can be applied to extract handcrafted features from these images. Among the most recently researched techniques, we can cite local binary pattern (LBP), grey level co-occurrence matrix (GLCM), speeded up robust features (SURF) or fractal geometry, which were applied for kidney tissue analysis (Simon et al., 2018), breast cancer classification (Yu et al., 2019), colon cell nuclei detection (Ama-lina et al., 2019) and lymphoma classification (Ribeiro et al., 2018), respectively. However, the main research focus for this area in recent years has been the application of deep learning approaches, more specifically, the use of CNNs.

The CNN models have shown to be efficient for the classification of objects, mainly in multiclass problems (Jiang & Su, 2018; Juefei-Xu et al., 2017), face recognition (Song et al., 2020) and satellite images (Akshay et al., 2020). However, these relevant results are not as often in the context of histological images (Araújo et al., 2017; Cireşan et al., 2013; Wang et al., 2014). One of the reasons is that CNNs require large sets for training, given that a major part of the public histological datasets available contains a limited number of samples (Bošnački et al., 2019; Yamashita et al., 2018). To handle this situation, more data is generated for training by applying rotation, mirroring or region cutting on the images. Nonetheless, this data augmentation raises, even more, the high computational cost of a CNN (Liu et al., 2019).

One of the possible solutions to reduce processing time consists in simplifying the network architecture by reducing the number of layers. However, the removal of deeper layers may hinder the image analysis from a global perspective (Araújo et al., 2017), which may compromise the network performance. Some alternative approaches, like hybrid networks, have been explored. These approaches associate non-deep learning techniques such as Gabor filters or LBP operators with the convolution operations of a CNN, which allows replacing some of the network's layers (Jiang & Su, 2018; Juefei-Xu et al., 2017). Other approaches aim to achieve a lower processing time by reducing the images' dimensionality. In Kausar et al. (2019), the authors applied Haar-wavelet decomposition on breast histology images and used the decomposed images as input to a CNN.

Recent researches have shown that a fusion of handcrafted features with deep learning models can enhance common approaches (Nanni et al., 2017). The application of fractal features, which have provided relevant results in different contexts of histological image classification such as breast tumours (Roberto et al., 2019), colorectal tumours (Ribeiro et al., 2019) and NHLs (Roberto et al., 2017), could also be associated to hybrid CNNs. In Xu et al. (2017), CNNs were applied to extract values from an invariant fractal dimension filter for detecting object curves in greyscale images. The authors in Mohammed et al. (2018) applied multifractal analysis to quantify and detect breast cancer, classifying the generated feature vectors using deep learning. However, an approach similar to the proposed by Kausar et al. (2019), wherein the CNN receives as input secondary images generated by a specific technique, has not yet been experimented in the fractal geometry context. Moreover, methods that directly associate fractal geometry with CNNs through an ensemble for the classification of histological images were not found in the literature.

In this paper, we propose a novel approach, which we name as Fractal Neural Network (FNN), to classify histological images through the association of fractal geometry and CNNs. In our proposal, fractal features are extracted from the histology images and then rearranged to generate an artificial RGB feature image. Both this artificial image and the correspondent original image are given as input to a CNN

ensemble, wherein a classification based on the sum rule outputs the class prediction. This new method provides the following contributions to the literature:

1. Double-CNN classification ensemble wherein an image generated from handcrafted fractal features and the respective regular image are given as input to a CNN;
2. An adaptive method that is able to classify different sets of histological images, including datasets with imbalanced classes, few samples and varying image dimensions;
3. The combination of different fractal measures to provide a set of features capable of describing the image's properties;
4. A deep learning model that requires a small number of training epochs, even when classifying new types of histology images.

In the second section of this paper, we provide a technical background on the use of fractal geometry for feature extraction from colour images. The proposed methodology is presented in Section 3, and in Section 4, the results obtained by applying the method on the tested datasets are presented and discussed. Finally, we conclude the paper in Section 5, with an overview of the obtained results and suggestions for future researches.

2. Technical background

2.1. Fractal features

Fractal geometry is a concept designed for the study of shapes that could not be defined by Euclidean geometry (Mandelbrot, 1975). Shapes present in nature such as a coastline, clouds, trees or lightning are examples of structures that do not have well-defined patterns. With fractal-based approaches, these structures can be represented by observations through different scales. In computer vision, such techniques are known as multiscale. Among the most common ones, we can highlight the box-counting (Nikolaidis et al., 2011) and the gliding-box (Ivanovici & Richard, 2011) algorithms. The application of these algorithms consists in splitting the images onto different scales and then extracting features from each sub-image. For the representation of numerical features using fractal approaches, we have the fractal dimension (FD), lacunarity (LAC) and percolation (PERC) as three of the most relevant. A multiscale and multidimensional analysis of the image are performed in order to obtain these features.

One of the approaches available in the literature for multiscale analysis is the gliding-box algorithm (Ivanovici & Richard, 2011). The main advantage of this approach is that it can be applied on datasets containing images with different resolutions since the output features are given in relation to the scale instead of being absolute values. This algorithm consists in placing a box β_i sized $L \times L$ on the left superior corner of the image, wherein L is given in pixels. This box glides through the image, one column and then one row at a time. After reaching the end of the image, the box is repositioned at the starting point and the value of L is increased by 2. On an image sized $H \times W$, the total number T of boxes β_i for a scale L is given by Eq. (1):

$$T(L) = (H - L + 1) \times (W - L + 1) \quad | \quad L \leq \min(H, W). \quad (1)$$

For each time the box β_i is moved, a multidimensional analysis of colour similarity is performed for every pixel inside it. This is done by assigning the centre pixel to a vector $f_c = r_c, g_c, b_c$, where r_c, g_c and b_c correspond to the colour intensities for each of the RGB colour channels of given pixel. The other pixels in the box are assigned to a vector $f_i = r_i, g_i, b_i$ and compared to the centre pixel by calculating a colour distance Δ . On the proposed approach, the Chessboard (Δ_h), Euclidean (Δ_e) and Manhattan (Δ_m) distances are calculated according to Eqs. (2)–(4).

$$\Delta_h = \max(|f_i(k_i) - f_c(k_c)|), k \in r, g, b. \quad (2)$$

Table 1
Structure of the probability matrix.

	3	5	...	L_{max}
1	$P(1, 3)$	$P(1, 5)$	...	$P(1, L_{max})$
2	$P(2, 3)$	$P(2, 5)$	...	$P(2, L_{max})$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
L^2	$P(L^2, 3)$	$P(L^2, 5)$	...	$P(L^2, L_{max})$

$$\Delta_e = \sqrt{\sum_k (f_i(k_i) - f_c(k_c))^2}, k \in r, g, b. \quad (3)$$

$$\Delta_m = \sum_k |f_i(k_i) - f_c(k_c)|, k \in r, g, b. \quad (4)$$

If the value of Δ corresponding to the distance between f_i and f_c is less than or equal to the scale L , then f_i is labelled as 1, otherwise f_i receives the label 0. An example of the pixels' labelling when a distance Δ is calculated for a box sized 3×3 is illustrated on Fig. 1.

This procedure converts a box that contains RGB values to one containing binary values. After performing this conversion for every box of every given L scale, a structure known as probability matrix is generated. Each element of the matrix corresponds to the probability P that m pixels on a scale L are labelled as 1 on each box. On Table 1, the visual representation of such matrix is presented. The matrix is normalized in a way that the sum of the elements in a column is equal to 1, as showed on Eq. (5).

$$\sum_{m=1}^{L^2} P(m, L) = 1, \forall L. \quad (5)$$

Noteworthy here that the probability matrix does not have the shape of an ordinary rectangular matrix, as the number of rows grows exponentially for each value of L . After the matrix is complete, the FD and LAC local values can be obtained.

2.1.1. Fractal dimension

FD is the most common technique to evaluate the fractal properties of an image. This is a measure for evaluating the irregularity and the complexity of a fractal. To obtain local FD features from the probability matrix, for each value of L , the FD denominated $D(L)$ is calculated according to Eq. (6):

$$D(L) = \sum_{m=1}^{L^2} \frac{P(m, L)}{m}. \quad (6)$$

2.1.2. Lacunarity

LAC is a measure complementary to FD and allows to evaluate how the space of a fractal is filled (Ivanovici & Richard, 2009). From the probability matrix, first and second-order moments are calculated with the Eqs. (7) and (8).

$$\mu(L) = \sum_{m=1}^{L^2} m P(m, L). \quad (7)$$

$$\mu^2(L) = \sum_{m=1}^{L^2} m^2 P(m, L). \quad (8)$$

The LAC value for a scale L is given by $\Lambda(L)$, which is obtained according to Eq. (9):

$$\Lambda(L) = \frac{\mu^2(L) - (\mu(L))^2}{(\mu(L))^2}. \quad (9)$$

Table 2
Summary of the obtained local features.

FD	LAC	PERC		
$D(3)$	$\Lambda(3)$	$C(3)$	$Q(3)$	$M(3)$
$D(5)$	$\Lambda(5)$	$C(5)$	$Q(5)$	$M(5)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$D(L_{max})$	$\Lambda(L_{max})$	$C(L_{max})$	$Q(L_{max})$	$M(L_{max})$

2.1.3. Percolation

PERC is a physical phenomenon that consists of the study of fluid properties on a porous media (Ghanbarian et al., 2015). Such media is said to be percolating if fluid can flow through the whole system, from the top to the bottom. In computer vision, this concept can be applied to verify the image porosity or some cluster properties regarding pixel neighbourhoods (Roberto et al., 2017). The first steps to obtain percolation features from a coloured image follow the same procedures described for obtaining the FD and LAC features. After calculating Δ , the generated binary matrices are given as input to a cluster labelling algorithm. Groups of nearby pixels that satisfied the criterion of the Δ distance are labelled to count the number of clusters on the image, as illustrated in Fig. 2. The symbol * indicates pixels labelled as 1.

Let c_i be the number of clusters on a box β_i , the feature $C(L)$ that represents the average number of clusters per box on a scale L is given by Eq. (10).

$$C(L) = \frac{\sum_{i=1}^{T(L)} c_i}{T(L)}. \quad (10)$$

Another feature that can be obtained is the average coverage area of the largest cluster in a box and is given by $M(L)$. Let m_i be the size in pixels of the largest cluster of the box β_i . The feature $M(L)$ is given according to Eq. (11):

$$M(L) = \frac{\sum_{i=1}^{T(L)} \frac{m_i}{L^2}}{T(L)}. \quad (11)$$

We can also verify whether a box β_i is percolating. This can be achieved due to a property that states a percolation threshold for different types of structures. In squared matrices (digital images), this threshold has the value of $p = 0.59275$ (Bird & Perrier, 2010), which means that if the ratio between pixels labelled as 1 and pixels labelled as 0 is greater or equal than p , the matrix is considered as percolating. Let Ω_i be the number of pixels labelled as 1 in a box β_i with size $L \times L$, we determine whether such box is percolating according to Eq. (12):

$$q_i = \begin{cases} 1, & \frac{\Omega_i}{L^2} \geq 0.59275. \\ 0, & \frac{\Omega_i}{L^2} < 0.59275. \end{cases} \quad (12)$$

This results in a binary value for q_i , wherein 1 indicates that the box is percolating. The feature $Q(L)$ regards the average occurrence of percolation on a scale L and can be obtained by Eq. (13):

$$Q(L) = \frac{\sum_{i=1}^{T(L)} q_i}{T(L)}. \quad (13)$$

The number of obtained local features depends on the total of observation scales L . Considering that L ranges from 3 to L_{max} with an increment of 2, the amount of local features corresponds to $5 \times (\frac{L_{max}-3}{2} + 1)$ for each Δ . A summary of these features is shown in Table 2.

2.2. Convolutional neural networks

Convolutional neural networks are a special type of deep learning model that learns features from low- and high-level patterns on grid-shaped data (Yamashita et al., 2018). The core of a CNN is usually built from three types of layers: convolution; pooling; and fully connected layers. While the first two perform feature extraction, the later classifies

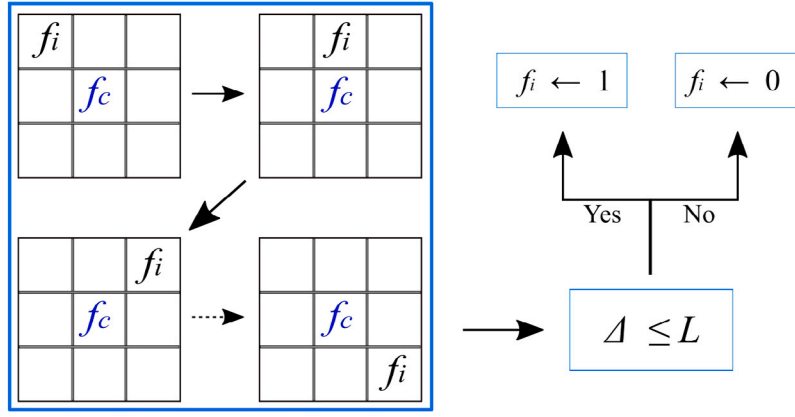


Fig. 1. Labelling the pixels on a 3×3 by calculating a distance Δ .

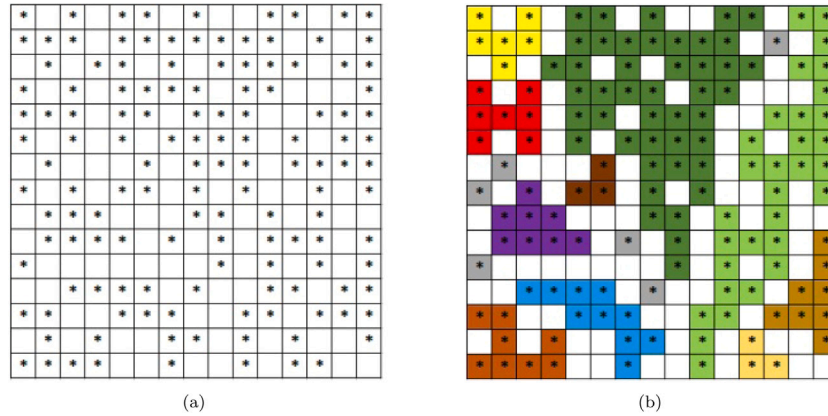


Fig. 2. Matrix before (a) and after (b) the application of a cluster labelling algorithm.

these features and usually outputs a label to be assigned to the input data.

The convolution layers are the base structures of the network, hence the name CNN. These layers usually consist of a series of two operations. The first is convolution, a simple linear procedure that performs element-wise product between the input data and small arrays of numbers called kernels, which are the only learnable parameters in this type of layers. The second operation consists of passing the convolution output through a non-linear activation function. Different functions have been applied, although the Rectified Linear Unit (ReLU), which is given by $f(x) = \max(0, x)$, became the most popular as it tends to reduce training time (Krizhevsky et al., 2012).

Pooling layers provide downsampling operations that reduce data dimensionality. This is usually done by selecting the maximum or sometimes the average value of an element in a patch and feeding it to the following layer. These pooling operations are performed in order not only to decrease the number of features but also to introduce a small invariance to translation and distortion of structures in the input data.

The fully connected layers consist of a series of one or more layers wherein every output is connected to every input of the following layer by a learnable weight. Usually, the last fully connected layer has the same number of nodes as the number of classes of the training dataset. In classification problems, its output corresponds to class probabilities, which are obtained by applying an activation function, such as softmax. The network's prediction is given by the class that obtained the highest probability value.

Table 3
Summary of the five tested datasets.

Dataset	Image	Classes	Samples	Resolution
UCSB	Breast tumours	2	58	896×768
CR	Colorectal tumours	2	165	From 567×430 to 775×522
NHL	Non-Hodgkin lymphoma	3	173	From 86×65 to 1388×1040
LG	Liver tissue	2	265	417×312
LA	Liver tissue	4	529	417×312

3. Methodology

3.1. Image datasets

We evaluated five histological image datasets represented by four different types of histological tissues: breast, colorectal, non-Hodgkin lymphoma and liver. One example from each type of histology image is presented in Fig. 3. An overview of all these datasets is presented in Table 3. In all five datasets, the tissue samples were stained with hematoxylin and eosin (H&E). For more details regarding the image's acquisition process, see Appendix A.

3.2. Method overview

The proposed approach is composed of two modules. The first module performs the extraction of local features by applying the fractal techniques described in Section 2.1. The output of this module consists of a set of 300 features, which were obtained by calculating the FD, LAC and PERC local values from each of the three distances Δ evaluated. The second module is composed of two CNNs whose goal is to perform classifications to obtain an array of probabilities.

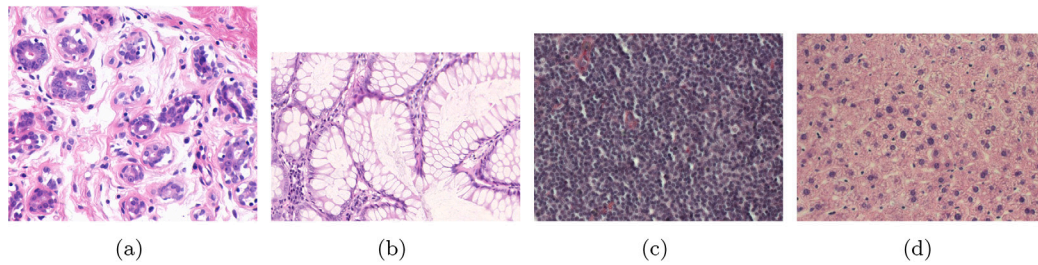


Fig. 3. Examples from a breast (a), colorectal, non-Hodgkin lymphoma (c) and liver (d) tissue image.

The input of the first CNN, henceforth named F-CNN, consists of an artificial image generated from the features extracted on the first module. The set of local features is reshaped into a $10 \times 10 \times 3$ RGB image, which is a procedure based on [Lumini and Nanni \(2018\)](#). The generated images are given as input to a CNN for classification. The second CNN, henceforth named O-CNN, receives as input the original image, wherein the class probabilities obtained from the classification of such image are summed to the respective class probabilities from the F-CNN. After this sum, the highest probability value indicates the class prediction. An overview of this approach is presented in [Fig. 4](#) and each step is described in details on the following sections.

3.3. Feature extraction module

The main stage of the proposed method consists in applying the techniques based on fractal geometry, described in [Section 2.1](#), on the images under investigation. FD, LAC, and PERC local features are extracted using multiscale and multidimensional approaches.

After being given as input to the FNN, the image is divided into different scales, according to the gliding-box algorithm. Each perceptron of the first layer represents a different scale with value L . This layer's function consists simply in generating a set of matrices for every region of the image and every assigned value of L , which ranges from 3 to 41 with an increment of 2. This will give a total of 20 features per function ($C(L)$, $Q(L)$, $M(L)$, $A(L)$ and $D(L)$), which results in 100 features for each Δ . In order not to generate distortion when resizing the image to give it as input to the CNNs, we chose this value for L_{max} as it generates a set of features which can be organized as a square matrix. Therefore, the 100 features obtained from $L_{max} = 41$ can be rearranged into a 10×10 matrix in order to provide a square image after applying the reshaping procedure. Moreover, this value was also chosen due to the relevant classification results obtained with a similar value ([Ribeiro et al., 2019](#)). The generated matrices are given as input to the second layer, which is a representation of the multidimensional approach of the methods described in [Section 2.1](#). In the proposed architecture, each perceptron performs the calculation of a different type of distance Δ between the pixels of the image.

The output of each perceptron in the second layer consists of a set of binary matrices, wherein the values labelled as 1 are pixels that matched the Δ criteria. These matrices are given as input to the perceptrons of the third layer, wherein the techniques described in [Section 2.1](#) for obtaining FD, LAC and PERC local values are finally applied. The resulting output consists of a set of 300 local features, which serve as input to the next module of the network. The original and fractal images are resized to match the required input dimensions before being given as input to both CNNs.

3.4. Classification module

On the proposed FNN, classification is performed by two CNNs. Both CNNs are fine-tuned on the deepest layer, as we applied transfer learning using models pre-trained on the ImageNet database ([Russakovsky et al., 2015](#)) to increase accuracy whilst reducing the training time.

Table 4
Selected pre-trained CNN models.

Model	Layers	Parameters	Input size
ResNet-50 (He et al., 2016)	50	2.6×10^7	$224 \times 224 \times 3$
ResNet-101 (He et al., 2016)	101	4.5×10^7	$224 \times 224 \times 3$
InceptionV3 (Szegedy et al., 2016)	48	2.4×10^7	$299 \times 299 \times 3$
Xception (Chollet, 2017)	71	2.3×10^7	$299 \times 299 \times 3$

3.4.1. Fractal features CNN — F-CNN

To serve as input for the incoming CNN classification, the feature vectors generated on the previous layers of the network must be converted into feature matrices. To do so, the 100 features obtained by each distance Δ are rearranged as a 10×10 matrix. The matrices generated by Δ_h , Δ_c and Δ_m correspond to the R, G and B colour channels, respectively. In [Fig. 5](#), one example of each tested dataset is shown to illustrate the reshaping procedure. Since each of the functions $C(L)$, $Q(L)$, $M(L)$, $A(L)$ and $D(L)$, obtained from a specific Δ , generate 20 features, each function is fit exactly into 2 columns of the matrix.

The generated images are given as input to the F-CNN, which outputs a score vector for each image indicating class probabilities.

3.4.2. Original images CNN — O-CNN

To fully explore the classification properties of a CNN, we chose to append a second CNN to the proposed architecture. The original images are given as input to this CNN and the class probabilities obtained from the output of the softmax layer are summed to the class probabilities obtained from the correspondent layer of the F-CNN where the images generated from fractal features were classified.

3.4.3. Transfer learning

To reduce training time and achieve good results with few training epochs we chose to use transfer learning instead of training both CNNs from scratch. Our proposed method applies network-based transfer learning ([Tan et al., 2018](#)), wherein the pre-trained network is partially reused and only the final layer is changed to match the number of classes. Therefore, we selected four CNN candidates that have provided relevant results in histology image classification recently ([Jiang et al., 2019](#); [Mahbod et al., 2018](#); [Rakhlin et al., 2018](#)) pre-trained on the ImageNet dataset. An overview of these four models is shown in [Table 4](#).

3.5. Performance evaluation

To obtain the best possible results with the proposed method and properly evaluate these, we applied a testing strategy divided into 4 stages.

Firstly, we evaluated which CNN model would be the most appropriate for the F-CNN and O-CNN slots in the proposed architecture. Then, we apply each of the CNNs on the 5 datasets with the number of training epochs ranging from 1 to 10. This verification aims to determine the smaller number of epochs needed to obtain the highest accuracies. After these experiments, the proposed approach is applied

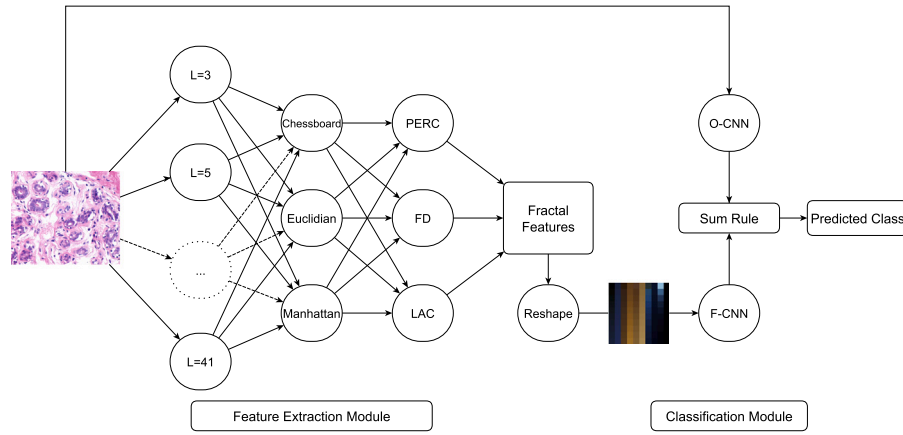


Fig. 4. Overview of the proposed FNN model.

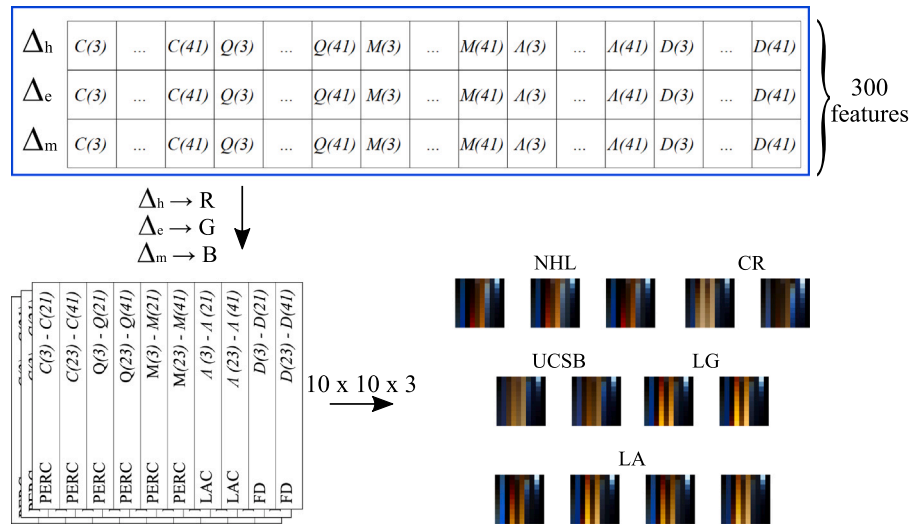


Fig. 5. Illustration of the procedure to rearrange the local features in order to create a RGB image.

to the datasets using the configuration results obtained from the previous tests. At last, we compare the proposed method with other common feature extraction techniques, analysing results obtained from using only the F-CNN as well as the ensemble with the O-CNN. To obtain these other features, we implemented the methods on Matlab R2019b, applied them on the same datasets and performed classification using the Rotation Forest algorithm available at the software Weka 3.6.13. We chose this classifier due to its relevant results obtained from other researches on histology image classification (Aličković & Subasi, 2017; Roberto et al., 2019).

All tests were performed on an Intel Xeon Silver 4116 CPU at 2.10 GHz with 128 GB of RAM and an NVIDIA GeForce RTX 2080Ti embedded, using Matlab R2019b. Since some of the tested datasets have a small number of samples, we chose to apply 10-fold cross-validation in all testing stages in order to avoid problems such as overfitting. The hyper-parameters used for training both CNNs were the *sgdm* solver; an initial learning rate of 0.01; a learning rate drop period of 2 epochs; a learning rate drop factor of 0.75; an L2 regularization value of 0.0001; a mini-batch size of 32; and the cross-entropy loss function.

4. Results and discussion

Before testing the proposed method on its complete implementation, it was necessary to determine the CNN models to be assigned to the F-CNN and O-CNN slots. We tested the performance of four of the most

Table 5

Loss evaluation of different CNN models and time performance in seconds.

Dataset		ResNet-50	ResNet-101	InceptionV3	Xception
NHL	F-CNN	0.510	0.494	0.478	0.385
	O-CNN	0.767	0.585	0.497	0.409
	Time	42.69	85.79	87.58	77.01
CR	F-CNN	0.350	0.312	0.339	0.269
	O-CNN	0.018	0.022	0.045	0.043
	Time	50.21	91.26	93.82	83.37
UCSB	F-CNN	0.567	0.741	0.621	0.556
	O-CNN	0.318	0.271	0.305	0.606
	Time	21.99	39.65	39.01	29.76
LG	F-CNN	0.163	0.118	0.151	0.118
	O-CNN	0.005	0.017	0.005	0.041
	Time	67.25	133.84	139.99	128.66
LA	F-CNN	0.175	0.189	0.196	0.193
	O-CNN	0.048	0.128	0.031	0.051
	Time	122.47	247.67	259.65	247.22

popular CNNs that have been applied to recent researches. In Table 5, the loss values for each dataset are presented as well as the average time in seconds required to train and classify the samples in 10 epochs.

We applied the Friedman non-parametrical test to verify whether the difference among the loss values were significant (Japkowicz & Shah, 2011). At $\alpha = 0.05$, we obtained $P_k = 0.2073$ for the F-CNN

Table 6

Results obtained from the application of the proposed method.

	Accuracy (%)	F-score
NHL	95.55	0.864
CR	99.39	0.994
UCSB	89.66	0.895
LG	99.62	0.996
LA	99.62	0.996
Avg	96.77	0.949
SD	4.334	0.058

and $P_k = 0.4144$ for the O-CNN, which indicates that there is not a significant difference when comparing the four tested CNN models. However, when comparing the time needed to perform training, the Friedman test indicated a significant difference ($P_k < 0.0001$) for all pairwise comparisons (Conover) involving the ResNet-50. Therefore, we chose the ResNet-50 to be assigned to both CNN slots of our proposed architecture not only due to its shorter training time but also due to the relevant results recently obtained in the classification of histology images (Mahbod et al., 2018; Rakhlin et al., 2018; Wang et al., 2019, 2020).

Then, we tested the performance of each of the two CNNs varying the number of training epochs. To prevent overfitting, and aiming to build a fast-training model, we chose to not go beyond 10 epochs for both CNNs. The results obtained from these tests, which can be observed in Appendix B, have shown that the F-CNN can provide relevant results after 4 epochs. On the other hand, the O-CNN presented significant performance values with only 2 training epochs, providing accuracies above 85% for all datasets, with an exception for the breast tumour images, due to its small number of samples. Nevertheless, the best results were obtained with 9 and 8 training epochs for the F-CNN and O-CNN, respectively. Therefore, these parameters were applied to the following tests.

We employed the proposed FNN using the configuration parameters obtained on the previous tests to evaluate the performance when applied to the five histology image datasets. The detailed results are shown in Table 6, including the average (Avg) and the standard deviation (SD) of all datasets.

These results show that the proposed method can perform well on the classification of histology images. With an exception for the UCSB dataset, accuracies above 95% were obtained, which can be an indicator of the method's adaptability to different categories of histological tissues. Despite dealing with imbalanced classes on all datasets, the proposed method was also able to provide F-Measure values above 0.850 in all cases. It is also noteworthy the excellent results obtained for the CR, LG and LA datasets, with performance values close to 1.0 for all evaluated metrics.

To verify how the proposed method fits among other computer vision approaches, we compare its performance with the results obtained by the LBP, Haralick, PERC, LAC and FD features. It is important to highlight that the fractal features used for comparison in this test consist of common feature vectors that are given as input to machine learning algorithms, which differs from our approach of reshaping the fractal features into an RGB image and feeding it to a CNN. Firstly, we compared the individual performance of these techniques without using any classification fusion or ensemble approach. Thus, we did not perform the ensemble scheme between the F-CNN and the O-CNN, as it is originally intended for our proposal. The results are shown in Table 7.

It is possible to note that without using any ensemble approach, the O-CNN, which consists simply of using a ResNet-50 for classifying the original images, provided the best results. An exception is made for the UCSB dataset, wherein the best results were provided by the PERC features. Moreover, apart from FD, the fractal features performed better than the LBP and Haralick descriptors in most cases. It can also be noted that the proposed F-CNN struggles to provide relevant results

Table 7

Accuracy values (%) obtained by different computer vision methods.

	NHL	CR	UCSB	LG	LA
LBP	72.83	67.27	79.31	80.75	71.46
Haralick	74.57	73.94	81.03	89.43	88.28
PERC	93.64	87.27	82.76	95.09	93.57
LAC	89.60	66.67	79.31	86.04	83.93
FD	78.03	59.39	62.07	58.49	50.47
F-CNN	82.66	86.06	79.31	95.09	93.57
O-CNN	93.64	98.79	81.03	98.87	98.68

Table 8

Accuracy values (%) obtained by the application of a classification ensemble between the O-CNN and other techniques.

	NHL	CR	UCSB	LG	LA
LBP	83.24	91.52	82.76	93.58	89.41
Haralick	87.28	92.12	82.76	95.47	92.25
PERC	94.22	97.58	86.21	99.62	99.24
LAC	94.22	98.18	86.21	99.62	98.87
FD	91.91	98.79	82.76	99.25	99.05
FNN	95.55	99.39	89.66	99.62	99.62

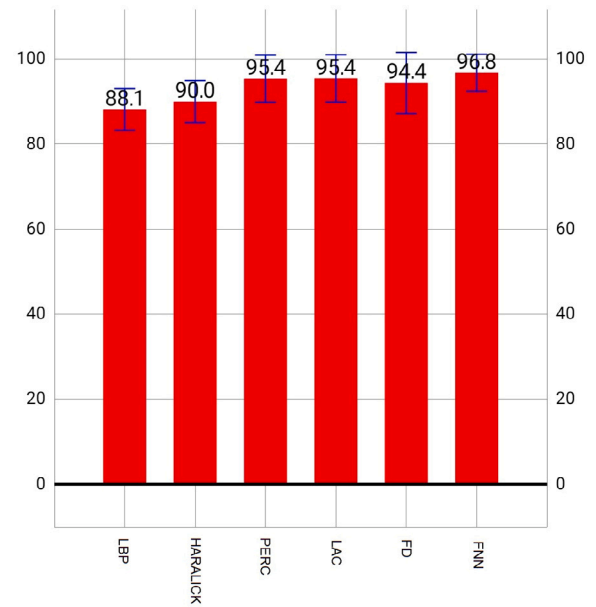


Fig. 6. Average accuracy (%) of the evaluated classification ensemble between the O-CNN and other techniques, applied to the five tested datasets.

when applied on its own. Thus, our method implemented an ensemble between the F-CNN and the O-CNN. However, since a comparison among an ensemble method and non-ensemble approaches would not be fair, we included the O-CNN classification to compose an ensemble with the other compared methods. These results are shown in Table 8.

It can be noted that merging the classification results of both CNNs not only improves the accuracy of the F-CNN but also enhances the performance of the O-CNN. When compared to the other techniques, the proposed method provided better accuracies in all datasets. The LG dataset was the only one where the results obtained by our method could be matched by other techniques. In this case, the same accuracy value of 99.62 was obtained by PERC, LAC and the FNN. Besides, a significant difference among the compared techniques was indicated according to the Friedman test ($P_k < 0.0001$ for $\alpha = 0.05$) in all pairwise comparisons (Conover). Our method also provided the highest average accuracy (96.8%) and the smallest standard deviation (4.334), as can be seen in the graph presented in Fig. 6.

An overview of the results obtained with the FNN in relation to other approaches in the context of histology image classification is

Table 9

Overview of the accuracy values (%) obtained by different approaches for NHL image classification.

Method	Approach	Accuracy
Bai et al. (2019)	GoogLeNet, intensity, texture and morphological features (DL+HC)	99.10
Jiang et al. (2018)	Morphology, entropy, GLCM and other handcrafted features (HC)	97.96
Martins et al. (2019)	FD and LAC (HC)	97.00
Janowczyk and Madabhushi (2016)	Caffe and AlexNet (DL)	96.58
FNN	ResNet-50, FD, LAC and PERC (DL+HC)	95.55
Roberto et al. (2017)	PERC (HC)	92.00
Ribeiro et al. (2018)	Colour normalization, PERC, LAC, GLCM and curvelet (HC)	86.14

Table 10

Overview of the accuracy values (%) obtained by different approaches for breast histology image classification.

Method	Approach	Accuracy
Li et al. (2019)	RefineNet and Atrous DenseNet (DL)	97.63
Yu et al. (2019)	CNN, LBP, SURF, GLCM and other handcrafted features (DL+HC)	96.67
Feng et al. (2018)	Stacked denoising autoencoder (DL)	94.41
Kausar et al. (2019)	Colour normalization, Haar wavelet decomposition and 16-layer CNN (DL)	91.00
FNN	ResNet-50, FD, LAC and PERC (DL+HC)	89.66
Roberto et al. (2019)	PERC (HC)	86.20
Papastergiou et al. (2018)	Spatial decomposition, tensors (DL)	84.67
Araújo et al. (2017)	Colour normalization, 13-layer CNN and SVM (DL)	83.30

shown in Tables 9–12. It can be noted that the methods that provided the best results on each classification task applied both deep learning (DL) and handcrafted (HC) features, except for breast image classification.

According to Table 9, we noted a lack of methods that used both HC features and DL for NHL image classification, which indicates a demand for new research in the area, although the best accuracy for this dataset was reported by Bai et al. (2019), wherein both DL and HC approaches were applied. Breast cancer classification remains a challenging task in computer vision since few methods were able to obtain accuracies above 95% when classifying these types of images, as observed in Table 10. We believe that the main reason for this is related to the short number of samples available on public breast tumour datasets. The methods that obtained higher accuracies on this dataset applied data augmentation techniques based on rotation, mirroring and cropping, wherein they were able to generate and train 200,000 (Yu et al., 2019) and 896,000 (Li et al., 2019) image patches. For colorectal images classification, our method obtained the best performance in relation to other approaches, as seen in Table 11. The second, third and fourth methods ranked in our comparison are all based on CNN approaches, without the use of HC features (Dabass et al., 2019; Sena et al., 2019; Tavolara et al., 2019). We believe that our method may encourage new research on the combination of HC and DL features for colorectal images classification since few papers on literature have applied similar approaches. Lastly, we compared different papers that applied computer vision approaches for the classification of gender and age from liver tissue images, as seen in Table 12. This is a context that has not been as explored as cancer detection challenges. However, it has already provided relevant results with high accuracy rates.

Overall, despite providing relevant results, most of these methods were implemented for specific classification tasks. Few computer vision approaches were able to perform well on different histological image categories (Nanni et al., 2019; Roberto et al., 2019). Moreover, recent papers have presented new approaches to deal with handcrafted fractal features. Roberto et al. (2017) proposed an approach to obtain global PERC features from RGB images. Later, in Roberto et al. (2019), local PERC features were also used for image classification. In Ribeiro et al. (2019), the authors used different types of handcrafted features, including FD, LAC and PERC, to classify colorectal tumours. However, these methods applied only the Δ_h metric to extract the fractal features. Moreover, none of these approaches explored the contribution of these features along with CNN models, which were able to provide high accuracy rates in several CAD systems for histopathology tasks (Andrearczyk & Whelan, 2017; Bai et al., 2019; Li et al., 2019). We believed that an ensemble method that addresses both fractal geometry and deep

Table 11

Overview of the accuracy values (%) obtained by different approaches for colorectal histology image classification.

Method	Approach	Accuracy
FNN	ResNet-50, FD, LAC and PERC (DL+HC)	99.39
Dabass et al. (2019)	31-layer CNN (DL)	96.97
Tavolara et al. (2019)	GAN and U-Net (DL)	94.02
Sena et al. (2019)	12-layer CNN (DL)	93.28
dos Santos et al. (2018)	Sample entropy and fuzzy (HC)	91.39
Roberto et al. (2019)	PERC (HC)	90.90
Bentaieb and Hamarneh (2018)	U-Net and AlexNet (DL)	87.50
Awan et al. (2019)	Colour normalization, U-Net and GoogLeNet (DL)	85.00

learning, which is the core of our proposal, could be able to improve these results when applied to different histology datasets. Hence, we implemented the FNN, wherein we could verify that its performance is compatible with recently published methods.

5. Conclusion

In this paper, we proposed an approach (FNN) based on the ensemble of two CNNs, wherein one of these receives as input images generated from fractal features, to classify different categories of histological images. We have shown that our proposal was able to provide relevant results, with accuracies above 89%, for all tested histopathology challenges. Also, accuracies greater than 99% were obtained for three out of the five evaluated datasets (CR, LG and LA). Besides, we achieved these results with a training time shorter than the required for other approaches such as Kausar et al. (2019), Sena et al. (2019) and Yu et al. (2019) to obtain similar performances. Therefore, we believe our proposal contributes to this research area not only due to its adaptability to different types of histology tissue and relatively low computing cost but mainly due to the applied feature vectors reshaping concept that allows the combined use of fractal features and CNNs. This approach has not yet been deeply explored and could provide new insights into the combined power of handcrafted features and deep learning. Nevertheless, there is still room for improvement, especially regarding the UCSB dataset classification, wherein hyper-parameter tuning could play a major role to improve the model's accuracy.

For future works, we propose the application of the FNN on other types of histology tissue images, a study for optimizing the F-CNN and O-CNN parameters, the inclusion of fractal global features in the classification ensemble and a deeper analysis of the fractal features reshaping procedure, for instance, the evaluation of different ways to

Table 12

Overview of the accuracy values (%) obtained by different approaches for gender and age classification from liver histology images.

Method	Approach	Acc. (gender)	Acc. (age)
Nanni et al. (2019)	6 CNN models and handcrafted descriptors (DL+HC)	100.00	100.00
Di Ruberto et al. (2016)	Statistical analysis and texture features (HC)	100.00	100.00
FNN	ResNet-50, FD, LAC and PERC (DL+HC)	99.62	99.62
Andrearczyk and Whelan (2017)	Texture CNN (DL)	99.10	98.20
Watanabe et al. (2016)	GIST descriptors, PCA and LDA (HC)	93.70	88.40

arrange the features or generating images using a different 3-channel colour model. We also intend to study the performance of each of the three distances separately, focusing on improving the classification of the global features to enhance its contribution to the ensemble. This method can also be applied to non-medical images, in order to provide different insights regarding the use of fractal features for image classification.

CRedit authorship contribution statement

Guilherme Freire Roberto: Conceptualization, Methodology, Software, Validation, Writing - original draft. **Alessandra Lumini:** Resources, Data curation, Writing - review, Supervision. **Leandro Alves Neves:** conceptualization, Writing - review, Supervision. **Marcelo Zanchetta do Nascimento:** conceptualization, Writing - review, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Dataset details

The experiments on this paper were performed on five histological image datasets. The first one consists of the breast cancer dataset provided by the Center of Bio-Image Informatics from the University of California, Santa Barbara (UCSB) (Gelasca et al., 2008). This dataset consists of 58 breast tissue images split into two groups: benign (32) and malignant (26).

The second dataset (CR) consists of 165 colorectal tissue images (Sirinukunwattana et al., 2017), also split into benign (74) and malignant (91) tumours. To acquire the images, histological sections were digitally photographed with a Zeiss MIRAX MIDI Slide Scanner with a scaled pixel resolution of 0.620 μm , which corresponds to a magnification of 20x.

The third dataset (NHL) is composed by 173 non-Hodgkin lymphoma images divided into three classes: MCL — mantle cell lymphoma (99); FL — follicular lymphoma (62); and CLL — chronic lymphocyte leukaemia (12). For the acquisition of the images, a light microscope Zeiss AxioScope with a 20x objective and a coloured digital camera (AXio Cam MR5) were used. The obtained images were recorded without compression, with a resolution of 1388 $\times$ 1040 pixels, a 24-bit quantization ratio and the RGB colour model. Regions of interest were

Table B.13

Accuracy and average values (%) for the F-CNN with varying training epochs.

Epochs	NHL	CR	UCSB	LG	LA	Avg.
1	78.03	77.58	60.34	68.68	77.88	72.50
2	80.35	78.79	70.69	81.13	81.66	78.52
3	78.61	83.03	68.97	91.32	84.88	81.36
4	82.66	83.64	70.69	88.68	86.39	82.41
5	84.97	89.09	74.14	93.21	86.77	85.63
6	84.39	86.67	72.41	89.81	91.68	84.99
7	84.97	88.48	68.97	90.94	91.68	85.01
8	84.39	88.48	72.41	91.70	90.36	85.47
9	82.66	86.06	79.31	95.09	93.57	87.34
10	83.81	86.06	77.59	95.47	93.19	87.23

Table B.14

Accuracy and average values (%) for the O-CNN with varying training epochs.

Epochs	NHL	CR	UCSB	LG	LA	Avg.
1	83.82	93.33	56.90	98.49	92.82	85.07
2	87.28	98.18	58.62	98.49	95.27	87.57
3	89.02	98.79	68.97	98.49	96.03	90.26
4	90.75	98.78	72.41	98.11	96.68	91.33
5	90.75	100.00	75.86	99.62	98.11	92.87
6	88.44	99.39	79.31	99.62	97.16	92.79
7	90.17	99.39	74.14	99.62	98.49	92.36
8	93.64	98.79	81.03	98.87	98.68	94.20
9	89.60	98.18	82.76	98.87	99.05	93.69
10	88.44	99.39	75.86	98.82	99.43	92.40

later selected by a specialist (Tosta et al., 2017). This dataset was made publicly available by the National Cancer Institute and the National Institute on Ageing (Shamir et al., 2008).

The two following datasets were both provided by the Atlas of Gene Expression in Mouse Ageing Project (AGEMAP) and are composed by liver tissue obtained from mice (AGEMAP, 2020). The images were acquired by a Carl Zeiss Axiovert 200 microscope and 40x objective. All images have the same resolution of 417 $\times$ 312 pixels. The fourth dataset (LG) consists of 265 liver tissue images obtained from male (150) and female (115) mice on a caloric restriction diet. The fifth dataset (LA) consists of 529 images split into four classes, wherein each represents a different age group of female mice on ad libitum diets: one (100), six (115), 16 (162) and 24 (152) months old.

Appendix B. Accuracies from different training epochs

These are the results of the experiments to determine the most suitable number of training epochs for each of the CNNs. The data are shown in Tables B.13 and B.14 for the F-CNN and the O-CNN, respectively.

References

- AGEMAP, N. I. o. A. (2020). The atlas of gene expression in mouse ageing project (agemap). <https://ome.grc.nia.nih.gov/iicbu2008/agemap/index.html> Accessed: 05-04-2020.
- Akshay, S., Mytravarun, T., Manohar, N., & Pranav, M. (2020). Satellite image classification for detecting unused landscape using CNN. In *2020 International Conference on Electronics and Sustainable Communication Systems (ICESC)* (pp. 215–222). IEEE.
- Aličković, E., & Subasi, A. (2017). Breast cancer diagnosis using GA feature selection and rotation forest. *Neural Computing and Applications*, 28(4), 753–763.

- Amalina, N. N., Ramadhani, K. N., & Sthevanie, F. (2019). Nuclei detection and classification system based on speeded up robust feature (SURF). *EMITTER International Journal of Engineering Technology*, 7(1), 1–13.
- Andrearczyk, V., & Whelan, P. F. (2017). Deep learning for biomedical texture image analysis. In *Proceedings of the Irish Machine Vision & Image Processing Conference*. Irish Pattern Recognition & Classification Society (IPRCS).
- Araújo, T., Aresta, G., Castro, E., Rouco, J., Aguiar, P., Eloy, C., Polónia, A., & Campilho, A. (2017). Classification of breast cancer histology images using convolutional neural networks. *PLoS One*, 12(6).
- Awan, R., Al-Maadeed, S., Al-Saady, R., & Bouridane, A. (2019). Glandular structure-guided classification of microscopic colorectal images using deep learning. *Computers and Electrical Engineering*, Article 106450.
- Bai, J., Jiang, H., Li, S., & Ma, X. (2019). NHL pathological image classification based on hierarchical local information and googlenet-based representations. *BioMed Research International*, 2019.
- Bentaieb, A., & Hamarneh, G. (2018). Adversarial stain transfer for histopathology image analysis. *IEEE Transactions on Medical Imaging*, 37(3), 792–802.
- Bird, N., & Perrier, E. (2010). Multiscale percolation properties of a fractal pore network. *Geoderma*, 160(1), 105–110.
- Bošnački, D., van Riel, N., & Veta, M. (2019). Deep learning with convolutional neural networks for histopathology image analysis. In *Automated Reasoning for Systems Biology and Medicine* (pp. 453–469). Springer.
- Chollet, F. (2017). Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 1251–1258).
- Cirşan, D. C., Giusti, A., Gambardella, L. M., & Schmidhuber, J. (2013). Mitosis detection in breast cancer histology images with deep neural networks. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 411–418). Springer.
- Dabass, M., Vig, R., & Vashisth, S. (2019). Five-grade cancer classification of colon histology images via deep learning. In *Communication and Computing Systems: Proceedings of the 2nd International Conference on Communication and Computing Systems (ICCCS 2018)*, December 1–2, 2018 (p. 18). Gurgaon, India: CRC Press.
- Di Ruberto, C., Putzu, L., Arabnia, H., & Quoc-Nam, T. (2016). A feature learning framework for histology images classification. In *Emerging Trends in Applications and Infrastructures for Computational Biology, Bioinformatics, and Systems Biology: Systems and Applications* (pp. 37–48). Elsevier Press.
- dos Santos, L. F. S., Neves, L. A., Rozendo, G. B., Ribeiro, M. G., do Nascimento, M. Z., & Tosta, T. A. A. (2018). Multidimensional and fuzzy sample entropy (sampenmf) for quantifying h&e histological images of colorectal cancer. *Computers in Biology and Medicine*, 103, 148–160.
- Feng, Y., Zhang, L., & Yi, Z. (2018). Breast cancer cell nuclei classification in histopathology images using deep neural networks. *International Journal of Computer Assisted Radiology and Surgery*, 13(2), 179–191.
- Gelasca, E. D., Byun, J., Obara, B., & Manjunath, B. (2008). Evaluation and benchmark for biological image segmentation. In *IEEE International Conference on Image Processing*.
- Ghanbarian, B., Hunt, A. G., Skinner, T. E., & Ewing, R. P. (2015). Saturation dependence of transport in porous media predicted by percolation and effective medium theories. *Fractals*, 23(01), Article 1540004.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 770–778).
- Howlader, N., Noone, A., Krapcho, M., Miller, D., Bishop, K., Kosary, C., Yu, M., Ruhl, J., Tatalovich, Z., & Mariotto, A. (2017). *Seer cancer statistics review, 1975–2014* (pp. 1–12). Bethesda, MD: National Cancer Institute.
- Huang, Q., Zhang, F., & Li, X. (2018). Machine learning in ultrasound computer-aided diagnostic systems: a survey. *BioMed Research International*, 2018.
- Ivanovici, M., & Richard, N. (2009). The lacunarity of colour fractal images. In *Image Processing (ICIP), 2009 16th IEEE International Conference on* (pp. 453–456). IEEE.
- Ivanovici, M., & Richard, N. (2011). Fractal dimension of color fractal images. *IEEE Transactions on Image Processing*, 20(1), 227–235.
- Janowczyk, A., & Madabhushi, A. (2016). Deep learning for digital pathology image analysis: A comprehensive tutorial with selected use cases. *Journal of pathology informatics*, 7.
- Japkowicz, N., & Shah, M. (2011). *Evaluating learning algorithms: a classification perspective*. Cambridge University Press.
- Jiang, Y., Chen, L., Zhang, H., & Xiao, X. (2019). Classification of h&e stained breast cancer histopathology images based on convolutional neural network. In *Journal of Physics: Conference Series*, Vol. 1302. IOP Publishing, Article 032018.
- Jiang, H., Li, Z., Li, S., & Zhou, F. (2018). An effective multi-classification method for NHL pathological images. In *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (pp. 763–768). IEEE.
- Jiang, C., & Su, J. (2018). Gabor binary layer in convolutional neural networks. In *2018 25th IEEE International Conference on Image Processing (ICIP)* (pp. 3408–3412). IEEE.
- Jothi, J. A., & Rajam, V. M. A. (2017). A survey on automated cancer diagnosis from histopathology images. *Artificial Intelligence Review*, 48(1), 31–81.
- Juefei-Xu, F., Naresh Boddeti, V., & Savvides, M. (2017). Local binary convolutional neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 19–28).
- Kausar, T., Wang, M., Idrees, M., & Lu, Y. (2019). Hwdcnn: Multi-class recognition in breast histopathology with haar wavelet decomposed image based convolution neural network. *Biocybernetics and Biomedical Engineering*, 39(4), 967–982.
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems* (pp. 1097–1105).
- Li, Y., Xie, X., Shen, L., & Liu, S. (2019). Reverse active learning based atrous densenet for pathological image classification. *BMC Bioinformatics*, 20(1), 445.
- Liu, L., Chen, J., Fieguth, P., Zhao, G., Chellappa, R., & Pietikäinen, M. (2019). From bow to CNN: Two decades of texture representation for texture classification. *International Journal of Computer Vision*, 127(1), 74–109.
- Lumini, A., & Nanni, L. (2018). Convolutional neural networks for ATC classification. *Current Pharmaceutical Design*, 24(34), 4007–4012.
- Mahbod, A., Ellinger, I., Ecker, R., Smedby, Ö., & Wang, C. (2018). Breast cancer histological image classification using fine-tuned deep network fusion. In *International Conference Image Analysis and Recognition* (pp. 754–762). Springer.
- Mandelbrot, B. B. (1975). Les objets fractals: forme, hasard et dimension.
- Martins, A. S., Neves, L. A., Faria, P. R., Tosta, T. A., Bruno, D. O., Longo, L. C., & do Nascimento, M. Z. (2019). Colour feature extraction and polynomial algorithm for classification of lymphoma images. In *Iberoamerican Congress on Pattern Recognition* (pp. 262–271). Springer.
- Mohammed, M. A., Al-Khateeb, B., Rashid, A. N., Ibrahim, D. A., Ghani, M. K. A., & Mostafa, S. A. (2018). Neural network and multi-fractal dimension features for breast cancer classification from ultrasound images. *Computers and Electrical Engineering*, 70, 871–882.
- Mueller, J. L., Gallagher, J. E., Chitalia, R., Krieger, M., Erkanli, A., Willett, R. M., Gerads, J., & Ramanujam, N. (2016). Rapid staining and imaging of subnuclear features to differentiate between malignant and benign breast tissues at a point-of-care setting. *Journal of Cancer Research and Clinical Oncology*, 142(7), 1475–1486.
- Nanni, L., Brahnam, S., Ghidoni, S., & Maguolo, G. (2019). General purpose (genp) bioimage ensemble of handcrafted and learned features with data augmentation. *arXiv preprint arXiv:1904.08084*.
- Nanni, L., Ghidoni, S., & Brahnam, S. (2017). Handcrafted vs. non-handcrafted features for computer vision classification. *Pattern Recognition*, 71, 158–172.
- Nikolaidis, N., Nikolaidis, I., & Tsouros, C. (2011). A variation of the box-counting algorithm applied to colour images. *arXiv preprint arXiv:1107.2336*.
- Noone, A., Howlader, N., Krapcho, M., Miller, D., Brest, A., Yu, M., Ruhl, J., Tatalovich, Z., Mariotto, A., & Lewis, D. (2018). In K. Cronin (Ed.), *Seer Cancer Statistics Review, 1975–2015*. National Cancer Institute.
- Papastergiou, T., Zacharaki, E. I., & Megalooikonomou, V. (2018). Tensor decomposition for multiple-instance classification of high-order medical data. *Complexity*, 2018.
- Rakhlin, A., Shvets, A., Iglovikov, V., & Kalinin, A. A. (2018). Deep convolutional neural networks for breast cancer histology image analysis. In *International Conference Image Analysis and Recognition* (pp. 737–744). Springer.
- Ribeiro, M. G., Neves, L. A., do Nascimento, M. Z., Roberto, G. F., Martins, A. S., & Tosta, T. A. A. (2019). Classification of colorectal cancer based on the association of multidimensional and multiresolution features. *Expert Systems with Applications*, 120, 262–278.
- Ribeiro, M. G., Neves, L. A., Roberto, G. F., Tosta, T. A., Martins, A. S., & do Nascimento, M. Z. (2018). Analysis of the influence of color normalization in the classification of non-hodgkin lymphoma images. In *2018 31st SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI)* (pp. 369–376). IEEE.
- Roberto, G. F., Nascimento, M. Z., Martins, A. S., Tosta, T. A., Faria, P. R., & Neves, L. A. (2019). Classification of breast and colorectal tumors based on percolation of color normalized images. *Computers and Graphics*, 84, 134–143.
- Roberto, G. F., Neves, L. A., Nascimento, M. Z., Tosta, T. A., Longo, L. C., Martins, A. S., & Faria, P. R. (2017). Features based on the percolation theory for quantification of non-hodgkin lymphomas. *Computers in Biology and Medicine*, 91, 135–147.
- Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., & Bernstein, M. (2015). Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3), 211–252.
- Sena, P., Fiorelli, R., Faglioni, F., Losi, L., Faglioni, G., & Roncucci, L. (2019). Deep learning techniques for detecting preneoplastic and neoplastic lesions in human colorectal histological images. *Oncology Letters*, 18(6), 6101–6107.
- Shamir, L., Orlov, N., Eckley, D. M., Macura, T. J., & Goldberg, I. G. (2008). Ilicbu 2008: a proposed benchmark suite for biological image analysis. *Medical & Biological Engineering & Computing*, 46(9), 943–947.
- Simon, O., Yacoub, R., Jain, S., Tomaszewski, J. E., & Sarder, P. (2018). Multi-radial LBP features as a tool for rapid glomerular detection and assessment in whole slide histopathology images. *Scientific Reports*, 8(1), 2032.
- Sirinukunwattana, K., Pluim, J. P., Chen, H., Qi, X., Heng, P.-A., Guo, Y. B., Wang, L. Y., Matuszewski, B. J., Bruni, E., & Sanchez, U. (2017). Gland segmentation in colon histology images: The glas challenge contest. *Medical Image Analysis*, 35, 489–502.
- Song, A.-P., Hu, Q., Ding, X.-H., Di, X.-Y., & Song, Z.-H. (2020). Similar face recognition using the IE-CNN model. *IEEE Access*, 8, 45244–45253.
- Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., & Wojna, Z. (2016). Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 2818–2826).

- Tan, C., Sun, F., Kong, T., Zhang, W., Yang, C., & Liu, C. (2018). A survey on deep transfer learning. In *International Conference on Artificial Neural Networks* (pp. 270–279). Springer.
- Tavolara, T. E., Niazi, M. K. K., Arole, V., Chen, W., Frankel, W., & Gurcan, M. N. (2019). A modular cgan classification framework: Application to colorectal tumor detection. *Scientific Reports*, 9(1), 1–8.
- Tosta, T. A. A., Faria, P. R., Neves, L. A., & do Nascimento, M. Z. (2017). Computational method for unsupervised segmentation of lymphoma histological images based on fuzzy 3-partition entropy and genetic algorithm. *Expert Systems with Applications*, 81, 223–243.
- Wang, Y., Huang, F., Zhang, Y., Zhang, R., Lei, B., & Wang, T. (2019). Breast cancer image classification via multi-level dual-network features and sparse multi-relation regularized learning. In *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (pp. 7023–7026). IEEE.
- Wang, Y., Lei, B., Elazab, A., Tan, E.-L., Wang, W., Huang, F., Gong, X., & Wang, T. (2020). Breast cancer image classification via multi-network features and dual-network orthogonal low-rank learning. *IEEE Access*, 8, 27779–27792.
- Wang, H., Roa, A. C., Basavanthally, A. N., Gilmore, H. L., Shih, N., Feldman, M., Tomaszewski, J., Gonzalez, F., & Madabhushi, A. (2014). Mitosis detection in breast cancer pathology images by combining handcrafted and convolutional neural network features. *Journal of Medical Imaging*, 1(3), Article 034003.
- Watanabe, K., Kobayashi, T., & Wada, T. (2016). Semi-supervised feature transformation for tissue image classification. *PloS One*, 11(12).
- Xu, H., Yan, J., Persson, N., Lin, W., & Zha, H. (2017). Fractal dimension invariant filtering and its cnn-based implementation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 3491–3499).
- Yamashita, R., Nishio, M., Do, R. K. G., & Togashi, K. (2018). Convolutional neural networks: an overview and application in radiology. *Insights into Imaging*, 1–19.
- Yu, C., Chen, H., Li, Y., Peng, Y., Li, J., & Yang, F. (2019). Breast cancer classification in pathological images based on hybrid features. *Multimedia Tools and Applications*, 1–21.